POLICY BRIEF / NOVEMBER 2026

# Road to Safety

Canadian collision patterns and practical priorities for reducing road harm

## 2,050 people killed in 2024

Transport Canada reports 130,551 injured people and 1,848 fatal collisions in the same year. Casualties count people, while collision totals count events.

## When collisions happen matters

The archived 1999-2021 records show the largest collision volume in late afternoon. The fatal share among recorded casualty collisions was highest in the 02:00 hour.

## Use local evidence to select interventions

National comparisons identify questions for local review. In our corridor-level pilot (City of Toronto + Quebec MTQ), linking 7,673 KSI crashes to 14,410 AADT-monitored road segments (104.44B VKT) yields a pooled rate of 4.71 KSI per 100M VKT, with urban Major Arterials carrying 12.6x the rate of Expressways (65.5% involving Vulnerable Road Users).

## Four priorities

1. Design and evaluate lower-speed local streets

2. Audit rural corridors for roadway departures

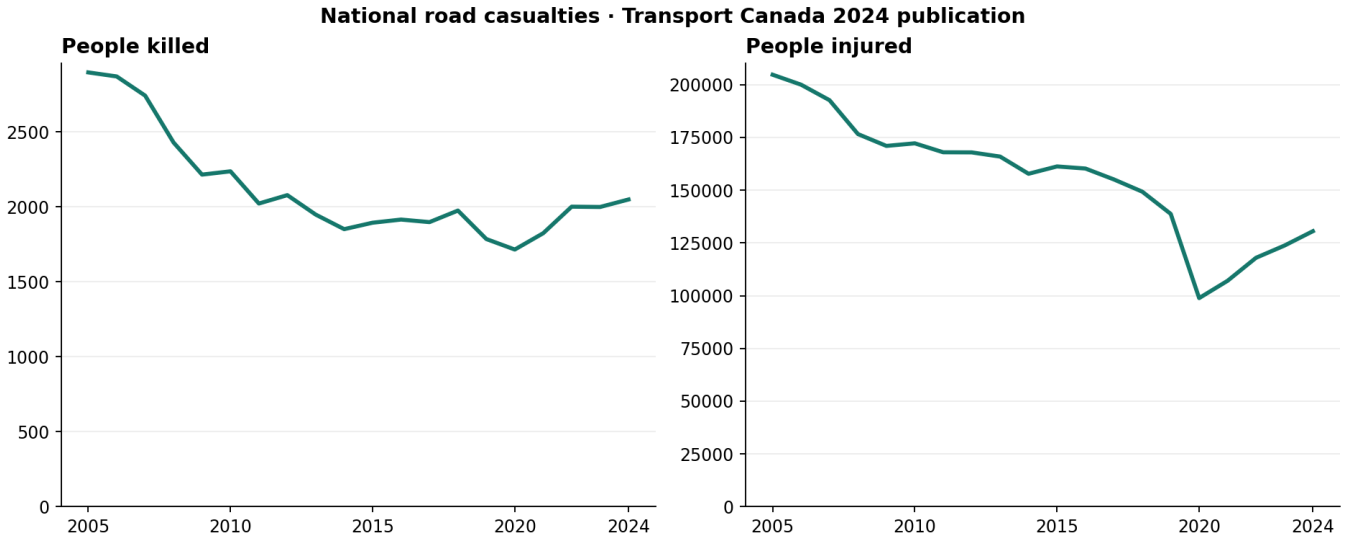
3. Audit visibility at crossings after dark

4. Link local exposure and collision reporting

Evidence periods. Archived event records 1999-2021 (2,949,846 collisions; P_USER and V_TYPE typologies integrated). Published national series 2005-2024. Toronto KSI + AADT corridor pilot 2006-2023.

# Two decades of road casualties

The annual publication tracks people killed and injured separately.



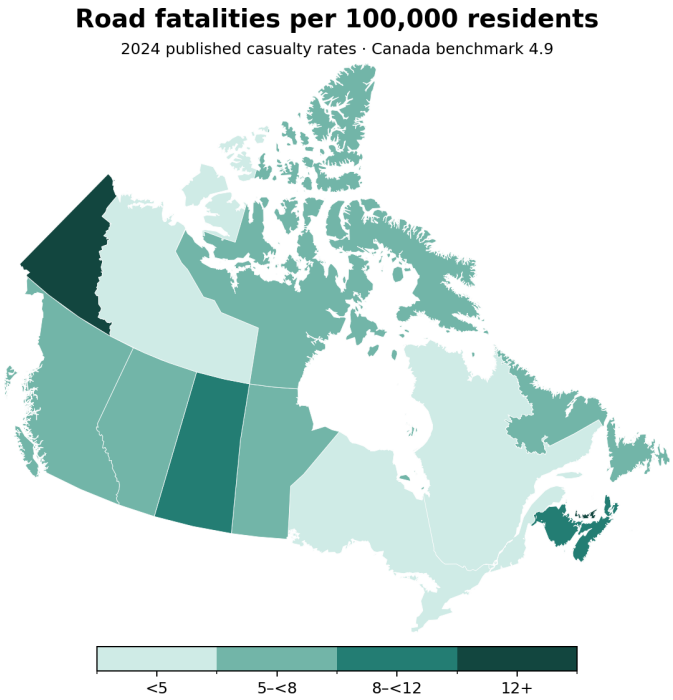

| Published 2024 measure | Count |
|---|---|
| Fatal collisions | 1,848 |
| Nonfatal injury collisions | 93,317 |
| People killed | 2,050 |
| People injured | 130,551 |

The series uses the Transport Canada 2024 publication vintage. The archived event releases used for the timing and severity analysis have separate coverage.

Transport Canada, Canadian Motor Vehicle Traffic Collision Statistics 2024

# Published rates vary across Canada

The Canada benchmark is 4.9 fatalities per 100,000 residents in 2024.



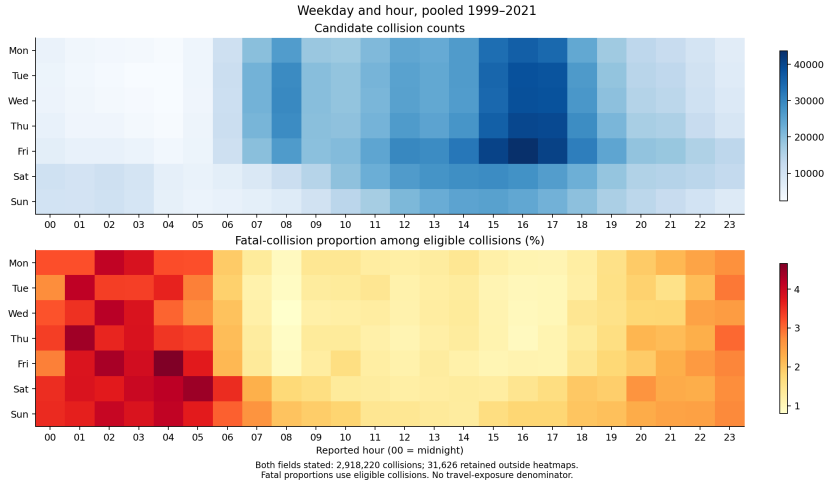

| Jurisdiction | Rate | Jurisdiction | Rate |
|---|---|---|---|
| Alberta | 5.9 | Nunavut | 7.2 |
| British Columbia | 5.2 | Ontario | 3.8 |
| Manitoba | 5.6 | Prince Edward Island | 12.0 |
| New Brunswick | 11.3 | Quebec | 4.2 |
| Newfoundland and Labrador | 7.1 | Saskatchewan | 8.4 |
| Northwest Territories | 4.4 | Yukon | 18.6 |
| Nova Scotia | 8.8 | Canada | 4.9 |

Ontario and Alberta figures are preliminary. New Brunswick figures are estimated. A Manitoba reporting change captured more minor injuries. Newfoundland and Labrador reflects reports processed by June 1, 2025. Rates in small populations can fluctuate sharply. Vehicle-kilometre estimates and resident-based rates answer different questions.

Statistics Canada 2021 cartographic boundaries

# Frequency and fatal share

The archived records describe outcomes among reported collisions with injury or death.




The 16:00 hour had the most reported collisions. At 02:00 the fatal share was highest among known hours. These patterns describe recorded collision outcomes. Time-specific travel exposure would be needed to compare occurrence rates.

## What the models can estimate

| Model / Operational Tier (2020-2021 Test) | Log loss | Fatal recall | Precision | Flagged |
|---|---|---|---|---|
| Training prevalence | 0.09746 | 100.0% | 2.0% | 100.0% |
| Logistic regression | 0.09257 | 82.3% | 2.9% | 55.8% |
| Random Forest | 0.09249 | 81.5% | 2.9% | 54.7% |
| LightGBM | 0.09150 | 81.9% | 3.0% | 53.6% |
| M2 Full Typology (P_USER + V_TYPE, Tier 1 only) | 0.09302 | 8.0% | 14.3% | 1.1% |
| M2 Full Typology (P_USER + V_TYPE, Tier 2 only) | 0.09302 | 34.5% | 7.3% | 9.4% |
| M2 Full Typology (P_USER + V_TYPE, Tier 3 only) | 0.09302 | 63.5% | 4.5% | 28.1% |

Reintroducing P_USER and V_TYPE typologies (M2) boosts 2020-2021 ROC-AUC from 0.696 to 0.763 and PR-AUC from 4.15% to 7.47%, isolating road tractors (OR 5.99x), unit heavy trucks (3.17x), pedestrians (2.69x) and heavy-freight x VRU crashes (12.5% fatal). Replacing the flat 80% recall threshold with a 3-Tier Operational Triage achieves 14.3% precision (14.2x relative risk) in Tier 1 while cutting Tier 3 false alarms in half.

The earlier calibration correction worsened probability scores in 2020 and 2021.

# Design for local conditions

Use the national findings to frame local questions and evaluate specific treatments.

## Design and evaluate lower-speed local streets

Transport Canada recorded 335 pedestrian fatalities in 2024. This national count identifies a burden and provides context for local evaluation. A Toronto quasi-experiment observed fewer pedestrian collisions after local limits fell from 40 to 30 km/h. The change relative to comparator streets was statistically uncertain.

**Action.** Municipal road authorities can test lower posted speeds and street designs that support them on suitable local streets with pedestrian activity.

**Evaluation.** Measure operating speeds and pedestrian injury collisions against comparable streets. Track pedestrian exposure and changes in traffic volume.

Fridman et al., Toronto quasi-experiment (2020)

## Audit rural corridors for roadway departures

The published 2024 table classifies 956 fatal collisions as rural and 855 as urban. The archived records also show a larger fatal share on curves with gradients than on straight and level roads. The FHWA implementation guidance identifies centerline and shoulder rumble strips as treatments for specific roadway-departure crash types.

**Action.** Provincial and territorial road authorities can use local crash records to select appropriate corridors and assess centerline or shoulder rumble strips. Review shoulder width, cycling access, pavement and nearby residences before installation.

**Evaluation.** Compare target fatal and injury crashes per vehicle-kilometre against similar untreated corridors over multiple years.

FHWA rumble strip implementation guidance

# Measure what changes

## Audit visibility at crossings after dark

Among reported casualty collisions with a stated hour, the fatal share is highest from 02:00 to 02:59. The national records lack site lighting and nighttime travel exposure for this comparison. FHWA lighting guidance recommends considering nighttime crash history, traffic and pedestrian activity when deciding where to improve crossing visibility.

**Action.** Municipal and provincial engineers can audit crossings and intersections with documented nighttime conflicts. Assess lighting, markings, glare and sight distance together.

**Evaluation.** Verify lighting performance and compare nighttime injury collisions at treated and comparable untreated sites, tracking pedestrian and vehicle volumes.

FHWA proven safety countermeasure on lighting

## Link local exposure and collision reporting

The archived collision releases lack location and exposure. Publication notes identify preliminary figures, estimates and differences in reporting practice. Canada's Road Safety Strategy 2035 and Beyond prioritizes collection, integration and timely sharing of high-quality data.

**Action.** Transport ministries, police and municipal data teams can agree on a versioned road segment key, coverage fields and harmonized outcome definitions. Preserve confidentiality through controlled access and aggregated releases.

**Evaluation.** Track geocoding completeness, matched exposure coverage, reporting delay and the share of severe outcomes with a verified definition.

CCMTA Road Safety Strategy 2035 and Beyond

## Data, corridor exposure and prospective drift governance

In Toronto's 104.44B-VKT corridor pilot, top Empirical Bayes priority corridors are Spadina Ave (13.98 EB KSI / 100M VKT), St Clair Ave W (11.26), Bay St (10.49), Danforth Ave (9.94), Bloor St W (8.41), Sherbourne St (8.19) and Yonge St (+19.03 excess KSI). For incoming 2022+ NCDB releases, the automated CKAN drift monitor enforces GREEN (O/E 0.92-1.08, PSI < 0.10), AMBER (intercept recalibration), and RED (quarantine / sentinel audit) gates.

NCDB records and dictionary / Transport Canada 2024 statistics / Statistics Canada boundary guide